12 CS/ISE/ASE/civil 42

Reg. No. | | | | | | | | | | |

[12EE42]

FOURTH SEMESTER B.E EXAMINATION, JUNE - 2014

ECOLOGY AND ENGINEERING ENVIRONMENT

Time: 1 Hour **Maximum Marks:40**

Answer all the questions. **40 x 1=40**

1. Which of the following statements about the forest is not correct?

a. Forest reduces soil erosion

b. Provides recreational opportunities

c. Provides economic development

d. None of the above

2. Forest and wild life are

a. Renewable resource

b. Non-renewable resources

c. Inexhaustible resources

d. None of these

3. Which of the following is not true about deforestation?

a. Population explosion is one of the reasons for deforestation.

b. Cleaning of forest for agriculture causes deforestation.

c. Deforestation is taking place only in developing courtiers.

d. Cash crop economy of third world is a cause of deforestation.

4. Deforestation generally decreases

a. Rainfall

b. Soil erosion

c. Draught

d. Global warming

5. Which of the following is not a viable protection against deforestation?

a. Reduce the consumption of forest and related products

b. Boycott products of companies involve in deforestation

c. Privatisation of forest land

d. Environmental education

6. Extensive planting of trees to increase cover is called

a. Afforestation

b. Agroforestation

c. Deforestation

d. Social forestry

7. Which of the following is responsible for desertification?

a. Deforestation

b. Mining

c. Overgrazing

d. All of the above

8. Forests prevent soil erosion by binding soil particles in their

a. Leaves

b. Roots

c. Buds

d. Stems

9. Many individuals of the same species living together in a defined area form a/an

a. Community.

b. Genus.

c. Population.

d. Ecosystem.

e. Variety.

10. Biological communities change because

a. Each stage modifies the environment and adapts for a later stage.

b. The soil is depleted and food gives out.

c. Old species move out and new species move in.

d. Old species evolve into new species.

e. New species move in, displacing old species.

11. Consider the following list of factors:

i. Dispersal

ii. Rainfall

iii. Mineral deposits

iv. Competition

v. Temperature.

Which of the factors listed above are most significant in determining the fact that belts of vegetation at successively higher altitudes often correspond to those at successively higher latitudes?

a. I and III

b. III and V

c. II and IV

d. II and V

e. I and V

12. The progressive series of changes that eventually produce a climax community on what was once a bare rocky island is an example of

a. Primary succession.

b. Speciation.

c. Secondary succession.

d. Evolution.

e. Eutrophication.

13. Some water, a small amount of soil, a few green aquatic plants and a fish were placed in a large bottle. The bottle was sealed to prevent the exchange of gases and other materials between its contents and the outside. The bottle was placed in a window to receive light during the daytime. Is carbon dioxide produced by the plants?

a. Yes, but it is produced only at night when the plants can no longer carry on photosynthesis.

b. Yes, it is produced all the time as a result of cellular respiration.

c. No, it is a waste product of animals only.

d. No, plants take in only the waste products exhaled by animals.

e. No, plants only produce oxygen.

14. The continuous movement of water from the oceans and freshwater sources to the air and land and back to the oceans is called the

a. nitrogen cycle

b. water cycle

c. runoff

d. oxygen cycle

15. During condensation,

a. water vapor turns to liquid

b. liquid turns to water vapor

c. liquid turns to solid

d. liquid falls to Earth as rain or snow

16. The polarity of water is due to

a. The charged ends of its molecules

b. The presence of minerals

c. Its purity

d. Its hardness

17. A man-made body of fresh water used as a source of drinking water is called a

a. aquifer

b. reservoir

c. water table

d. watershed

18 A land area in which surface runoff drains into a system of rivers is called a

a. aquifer

b. watershed

c. zone of aeration

d. zone of saturation

19. Incoming solar radiation is partly reflected back into space by the Earth's:

I. atmosphere

II. land surface

III. ocean surface

a. I only

b. II only

c. III only

d. II and III

e. I, II, and III

20. Which of the following is not a major greenhouse gas?

a. Carbon-dioxide

b. Methane

c. CFCs

d. Water vapor

e. Ozone

21. Kaveri water dispute is in between:

a. India and Pakistan

b. Punjab and Haryana

c. Uttar Pradesh and Madhya Pradesh

d. Karnataka and Tamil Nadu

22. The world as World Environmental day is celebrated on:

a. December 1

b. June 5

c. November 14

d. August 15

23. The first of the major environmental protection act to be promulgated in India was:

a. Water Act

b. Air Act

c. Environmental Act

d. Noise Pollution Rule

24. The Forest (Conservation) Act was enacted in the year:

a. 1986

b. 1974

c. 1980

d. 1972

25. The Forest (Conservation) Act extends to the whole of India except:

a. Uttar Pardesh

b. Karnataka

c. Jammu and Kashmir

d. Haryana

26. Which is the correct set of statements?

	Troposphere	Stratosphere
A.	Absorbs large proportions of long-wave radiation from above; temperature increases with altitude	Absorbs large proportions of short-wave radiation from below; temperature decreases with altitude
B.	Absorbs large proportions of long-wave radiation from below; temperature decreases with altitude	Absorbs large proportions of short-wave radiation from above; temperature increases with altitude
C.	Absorbs large proportions of short-wave radiation from above; temperature increases with altitude	Absorbs large proportions of long-wave radiation from below; temperature decreases with altitude
D.	Absorbs large proportions of short-wave radiation from below; temperature decreases with altitude	Absorbs large proportions of long-wave radiation from above; temperature increases with altitude

27. The ozone layer can be protected by

I. using substitutes for ozone-depleting chemicals.

II. reducing cattle production.

III. recycling old refrigerators.

a. I and II only

b. I and III only

c. II and III only

d. I, II and III

28. Which of the following has been most effective in reducing world emissions of CFCs?

a. Improved technologies in the development of electric cars

b. Aid programmes to developing countries

c. Policies to minimize the combustion of fossil fuels

d. International agreements, including the Montreal Protocol

29. What happens when sunlight interacts with pollutants in the lower atmosphere?

a. Destruction of the ozone layer occurs

b. Stratospheric ozone is produced

c. Thermal inversion occurs

d. Photochemical smog is produced

30. Which of the following pairs of statements about ozone is correct?

	Tropospheric ozone	Stratospheric ozone
A.	It is significantly affected by increasing levels of CFCs	It accumulates in high concentration to form the ozone layer
B.	It causes health problems in humans and animals	It is significantly affected by increasing levels of CFCs
C.	It accumulates in high concentration to form the ozone layer	It damages photosynthetic organisms
D.	It damages photosynthetic organisms	It causes health problems in humans and animals

31. Which of the following greenhouse gases are produced only by human activities?

a. Methane and chlorofluorocarbons (CFCs)

b. Carbon dioxide and water vapour

c. CFCs

d. Methane and water vapour

32. Problems with replacing CFCs (chlorofluorocarbons) with HFCs (hydrofluorocarbons) are that HFCs

I. are powerful greenhouse gases.

II. remain in the atmosphere much longer than CFCs.

III. are expensive to produce.

a. I and II only

b. II and III only

c. I and III only

d. I, II and III

33. Photochemical smog is formed when primary pollutants interact with which of the following?

a. Water vapour

b. Heat

c. Oxygen

d. Sunlight

34. Which of the following is most likely to lead to increased mutation rates in phytoplankton?

a. Increase in stratospheric ozone

b. Decrease in stratospheric ozone

c. Increase in tropospheric ozone

d. Decrease in tropospheric ozone

35. Which of the following are normally associated with the formation of photochemical smog?

I. Nitrogen oxides

II. Sunlight

III. Volatile organic compounds

IV. Clouds

a. I and III only

b. III and IV only

c. I, II and III only

d. I, II, III and IV

36. Which statement about the ozone layer is correct?

a. Fluorine released from chlorofluorocarbons combines with carbon dioxide molecules in the upper atmosphere to form ozone.

b. Acid rain is caused by an increase in the amount of ultraviolet radiation that penetrates the ozone layer.

c. The burning of fossil fuels causes the formation of ozone, which gradually diffuses to the stratosphere.

d. Ultraviolet radiation is absorbed during the process of stratospheric ozone formation

37. For which environmental impacts has acid deposition been identified as a significant cause?

I. Leaching of calcium from soil

II. Increasing the rate of respiratory problems in humans

III. Contamination of fish with toxic aluminium

IV. Increasing the productivity of marine phytoplankton

a. I and II only

b. I and IV only

c. I, II and III only

d. I, II, III and IV

38. Which of these measures would reduce the quantity of acid rain produced?

I. Removing SO2 from smokestacks

II. Switching to renewable energy

III. Using lime to neutralize acidified lakes

IV. Replacing coal with natural gas as energy source

a. I, II and IV only

b. II and IV only

c. I, II and III only

d. I, III and IV only

39. Which column in the table correctly shows the effects of the pollutant gas?

	A.	B.	C.	D.
	Sulfur dioxide	**Halogenated gases**	**Methane**	**Carbon dioxide**
increases the greenhouse effect	Yes	No	Yes	Yes
depletes stratospheric ozone	Yes	Yes	No	Yes
increases acidity of rain	Yes	No	No	Yes

40. Compressed Natural Gas (CNG) is:

a. Propane

b. Methane

c. Ethane

d. Butane